fn sample_geometric_exp_fast

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Proves soundness of `fn sample_geometric_exp_fast` in `mod.rs` at commit 0be3ab3e6 (outdated[1]). This proof is an adaptation of subsection 5.2 of [CKS20].

1 Hoare Triple

Precondition

Compiler-verified

Argument `x` is of type `RBig`, a bignum rational.

Caller-verified

$\texttt{x} \geq 0$

Pseudocode

```
def sample_geometric_exp_fast(x: RBig) -> int:
    if x == 0:
        return 0

    s, t = x.into_parts()

    while True:
        u = sample_uniform_ubig_below(t) #
        d = sample_bernoulli_exp(Rational(u, t)) #
        if d:
            break

    v = sample_geometric_exp_slow(1) #
    z = u + t * v
    return z // s
```

Postcondition

For any setting of the input parameter `x` such that the given preconditions hold, `sample_geometric_exp_fast` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as $\text{Geometric}(1 - \exp(-x))$.

2 Proof

Assume the preconditions are met.

[1] See new changes with `git diff 0be3ab3e6..7d1a2007 rust/src/traits/samplers/cks20/mod.rs`

Lemma 2.1. `sample_geometric_exp_fast` only returns `Err(e)` when there is a lack of system entropy.

Proof. `x` ≥ 0 and `x` is of type `Rbig`, so there exists some non-negative integer s and positive integer t such that $x = s/t$. This is why `Rational.into_numer_denom` is infallible. Since t is a positive integer, the preconditions on `sample_uniform_ubig_below` are met, `sample_uniform_int_below` can only return an error due to lack of system entropy, and u is a non-negative integer. Similarly, the preconditions on `sample_bernoulli_exp` and `sample_geometric_exp_slow` are met, and their definitions guarantee an error is only returned due to lack of system entropy. The only source of errors is from the invocation of these functions, therefore `sample_geometric_exp_fast` only returns `Err(e)` when there is a lack of system entropy. □

We now establish some lemmas that will be useful in proving the distribution of `out`.

Lemma 2.2. Conditioned on `sample_geometric_exp_fast` not returning `Err(e)`, the following hold:

- The value `u` sampled on line 8 is a realization of $U \sim \text{Uniform}(0, t)$ supported on $\{0, 1, \ldots, t-1\}$.
- Conditioned on a fixed realization u from line 8, the value `d` sampled on line 9 is a realization of $D \sim \text{Bernoulli}(\exp(-u/t))$.
- The value `v` sampled on line 13 is a realization of $V \sim \text{Geometric}(1 - \exp(-1))$.
- Moreover, U is independent of V, and the call on line 13 is independent of the loop outcome on lines 8–9.

Proof. Since `x` is an `RBig` rational and `x` ≥ 0, there exist integers $s \geq 0$ and $t > 0$ with $x = s/t$, so the argument t passed to `sample_uniform_ubig_below` on line 8 satisfies its preconditions. By the specification/definition of `sample_uniform_ubig_below`, the returned value `u` is uniformly distributed over $\{0, \ldots, t-1\}$, i.e., $U \sim \text{Uniform}(0, t)$.

On line 9, the call `sample_bernoulli_exp` is made with input `Rational(u,t)`. The preconditions of `sample_bernoulli_exp` are met (in particular, $u \in [0, t)$ and $t > 0$), and by its specification/definition, conditioned on the realized `u`, the returned bit `d` is distributed as $\text{Bernoulli}(\exp(-u/t))$.

On line 13, the call `sample_geometric_exp_slow` with argument 1 returns `v` distributed as $\text{Geometric}(1-\exp(-1))$ by its specification/definition.

Independence follows because line 13 is executed after the loop and uses fresh randomness. □

Lemma 2.3. [CKS20] Assume `u`, `d`, and `v` satisfy the conditions of Lemma 2.2. Conditioned on $d = \top$, if $z = u + t \cdot v$, then z is a realization of a random variable $Z \sim \text{Geometric}(1 - \exp(-1/t))$. Equivalently, $P[Z = z \mid D = \top] = (1 - e^{-1/t})e^{-z/t}$.

Proof. For any z, define $u_z := z \mod t$ and $v_z := \lfloor z/t \rfloor$, so that $z = u_z + t \cdot v_z$.

$$
\begin{aligned}
P[Z = z \mid D = \top] &= P[U = u_z, V = v_z \mid D = \top] && \text{since } z = u_z + t \cdot v_z \\
&= P[U = u_z \mid D = \top]\, P[V = v_z] && \text{as } U \text{ and } V \text{ are independent, and } V \text{ and } D \text{ are indepen} \\
&= \frac{P[U = u_z]}{P[D = \top]} P[D = \top \mid U = u_z] \cdot (1 - e^{-1})e^{-v_z} && \text{by Bayes' theorem} \\
&= \frac{1/t}{1/t \sum_{k=0}^{t-1} e^{-k/t}} e^{-u_z/t} \cdot (1 - e^{-1})e^{-v_z} && \text{since } P[D = \top] = \frac{1}{t}\sum_{k=0}^{t-1} e^{-k/t} \\
&= \frac{(1 - e^{-1})}{\sum_{k=0}^{t-1} e^{-k/t}} e^{-(u_z/t + v_z)} \\
&= \frac{(1 - e^{-1})}{(1 - (e^{-1/t})^t)/(1 - e^{-1/t})} \\
&= (1 - e^{-1/t})e^{-(u_z/t + v_z)} \\
&= (1 - e^{-1/t})e^{-z/t} && \text{since } z = u_z + t \cdot v_z
\end{aligned}
$$

□

Lemma 2.4. [CKS20] Fix $p \in (0, 1]$. Let G be a Geometric$(1-p)$ random variable, and $n \geq 1$ be an integer. Then $\lfloor G/n \rfloor$ is a Geometric$(1-q)$ random variable with $q = p^n$.

Proof.

$$\begin{aligned} P[\lfloor G/n \rfloor = k] &= P[nk \leq G < (k+1)n] && \text{any } G \text{ in the interval maps to } k \\ &= \sum_{l=kn}^{(k+1)n-1} (1-p)p^l \\ &= (1-p^n)p^{nk} \\ &= (1-q)q^k \end{aligned}$$

□

Lemma 2.5. If the outcome of `sample_geometric_exp_fast` is `Ok(out)`, then `out` is distributed as Geometric$(1 - \exp(-x))$.

Proof. By Lemma 2.2, `u`, `d`, and `v` have the distributions (and independence) required to apply Lemma 2.3. By Lemma 2.3, `z` is a realization of $Z \sim$ Geometric$(1 - \exp(-1/t))$; hence by Lemma 2.4, $\lfloor Z/s \rfloor$ is distributed as Geometric$(1 - \exp(-s/t))$. Finally, `out` $= \lfloor Z/s \rfloor$ and $s/t = x$, so `out` is distributed as Geometric$(1 - \exp(-x))$. □

Proof. 1 holds by 2.1 and 2.5. □

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.